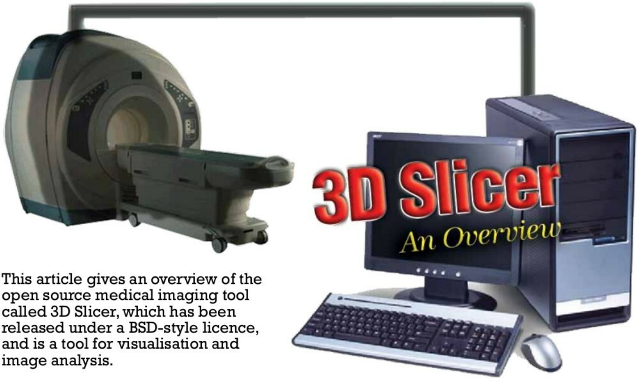

This article gives an overview of the open source medical imaging tool called 3D Slicer, which has been released under a BSD-style licence, and is a tool for visualisation and image analysis.

The computer has become an integral part of our lives and the medical field is no different. All the data collected by MRI scanners and instruments for endoscopy, thermography, etc, is analysed by software that employs heavy digital-signal-processing algorithms. Many open source tools have now begun to replace their proprietary counterparts in this domain, reducing the overall cost of operations to some extent.

3D Slicer is one such open source tool that is used for 3D image analysis and visualisation of the data collected by medical imaging instruments. Slicer provides a GUI with the data. Besides creating 3D surface models for conventional MRI scanned data, it has been used for non-rigid image segmentation and to incorporate models of neuro-vascular bundles (nerves, arteries, veins and lymphatics that run together in the body).

The origins of Slicer go back to 1998, when it was started just as a post-graduate thesis project between the Surgical Planning Laboratory at the Brigham and Women's Hospital, and the MIT Artificial Intelligence Laboratory. The latest release is Slicer 4, which is a Qt enabled version. Slicer is a cross-platform tool and can run on Linux, Windows, Mac, etc. It is intended to work with tomographic data of all types, particularly CT and MRI.

I remember the early days of the CT revolution, when the equipment was expensive, requiring $10,000 worth of software and around $4,000 for hardware. 3D Slicer, being open source, is completely free and has cut down a big chunk of that expense.

Requirements

The basic requirements to run Slicer on your PC are:

1) Windows XP or more recent, Linux (x86 or x86_64), Mac OS X (PPC or Intel)
2) 2 GB RAM (minimum)
3) At least 128 MB of on-board graphic memory

Mac OS X will need X11 or XQuartz to be installed prior to Slicer's installation.

Starting up

At the time of writing this article, I used release 3.6.3 and I ran it on Fedora 16 with no problems. You can get it from *http://www.slicer.org/pages/Special:SlicerDownloads*. As the *root* user, extract the zip file to */opt*, go to the directory where it has been extracted and fire up Slicer as follows:

```
tar -xvzf Slicer3-3.6.3-2011-03-04-linux-x86.tar.gz -C /opt
cd /opt/Slicer3-3.6.3-2011-03-04-linux-x86
./Slicer3
```

Try it out!

So, that's enough of talk! Let's try out an example of this amazing tool. First, you'll need raw data. Don't panic! You don't have to get an MRI scanner. For mere practice, you can easily download an example data set from: *http://bit.ly/MfYmnc*

Extract the downloaded zip file to any suitable location and give it a name like 'CT Data'. In Slicer, click *File → Add Volume* and then select the first file from 'CT Data'. You will see your data. It may take some time if your system is slow. Slicer automatically loads the three orthogonal slices.

The fourth last button on the toolbar (located at the top) can be used to re-size the image. Click on it and select 'Four-up layout' and the orientation will change as shown in Figure